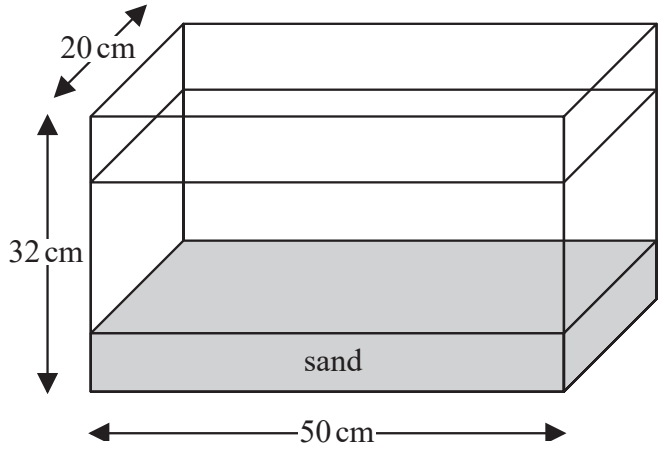
2 The diagram shows a fish tank in the shape of a cuboid.



The dimensions of the tank are 50 cm by 32 cm by 20 cm.

The tank is $\frac{3}{4}$ full of water and sand.

The ratio of the volume of water to the volume of sand is 5 : 1

Work out the number of litres of water in the tank.
You must show all your working.

Full tank: $50 \times 32 \times 20 = 32000 cm^3$.

1 mark
Volume of full tank

Water + sand: $\frac{3}{4} \times 32000 = 24000 cm^3$.

1 mark
Three quarters full

	Total	Water
	6	5
×4000	24000	20000

×4000

Water is 5 of 6 parts.

$24000 \div 6 \times 5 = 20000 cm^3$.

1 mark
Split in ratio 5:1

	ml	L
	1000	1
×20	20000	20

×20

1 mark
$1 cm^3 = 1 ml$.

$20000 \div 1000 = 20$ litres.

1 mark
20 litres

.....litres

(Total for Question 2 is 5 marks)